



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Ratings

Key Data Dimensions

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Review Rating, Shipping Type

Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for structure and statistics.

02

Clean & Standardize

Imputed missing Review Ratings with category median; renamed columns to snake_case.

03

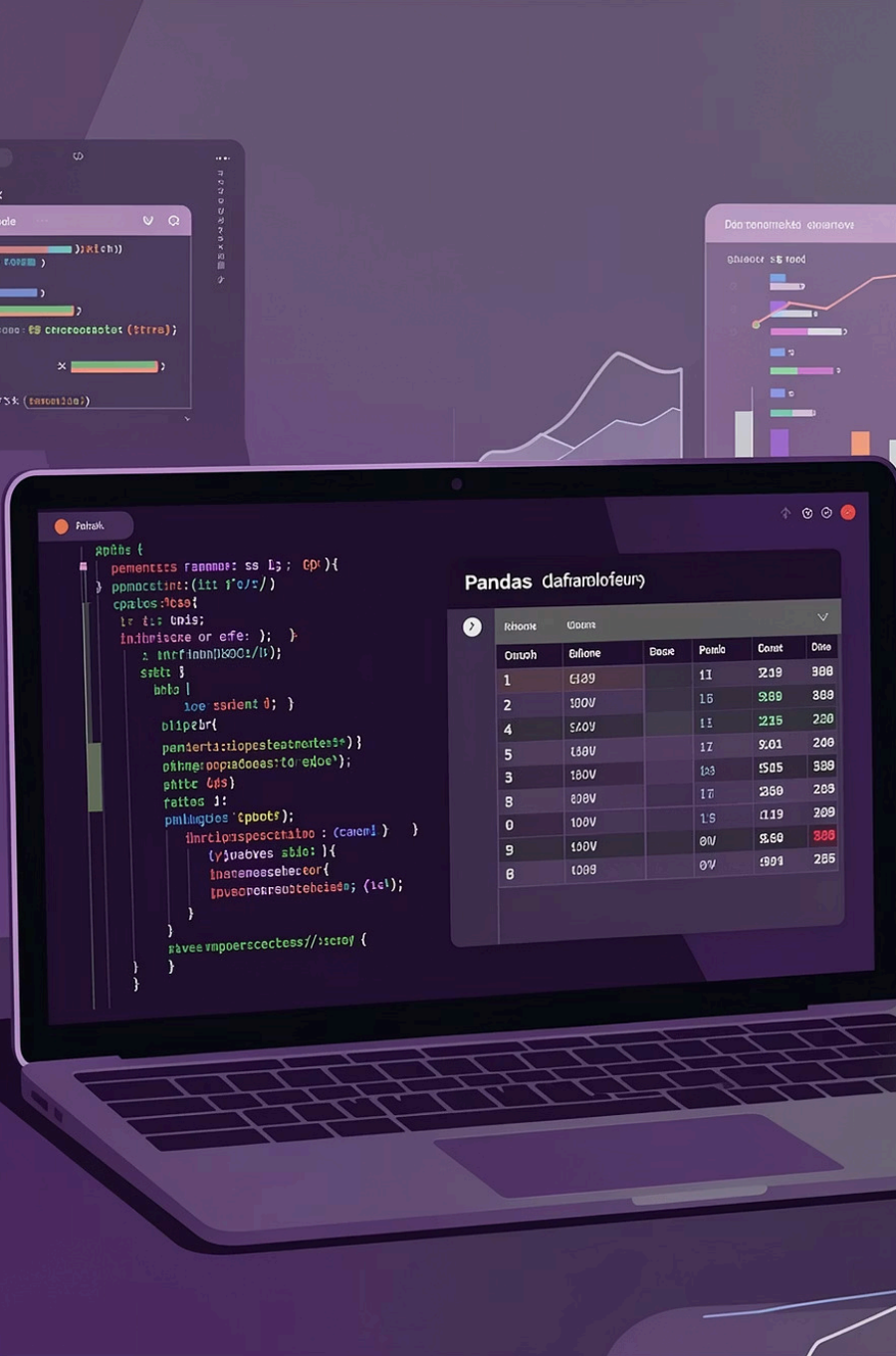
Feature Engineering

Created `age_group` and `purchase_frequency_days` columns; dropped redundant `promo_code_used`.

04

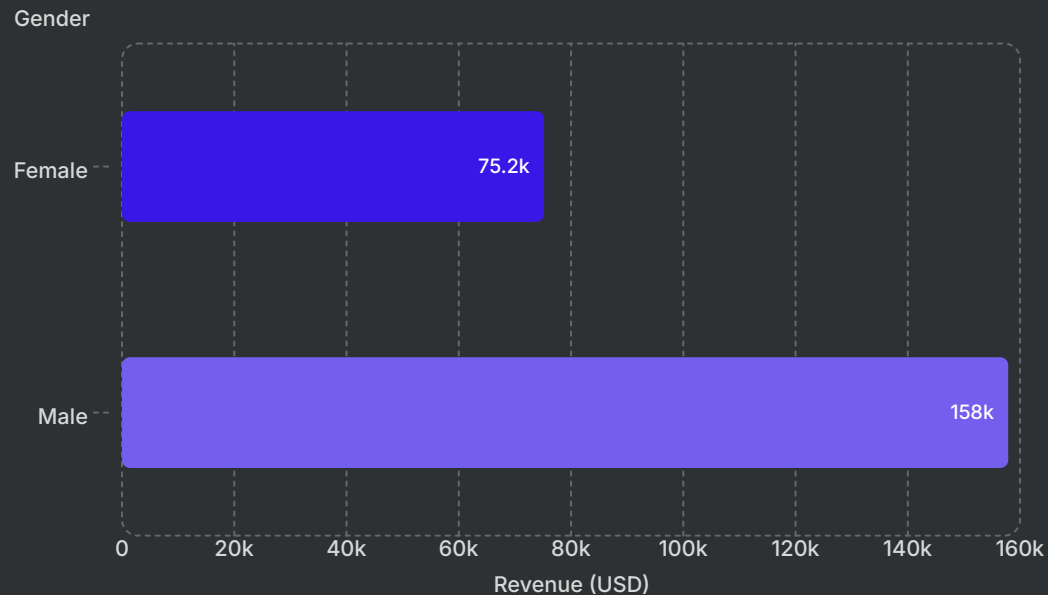
Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

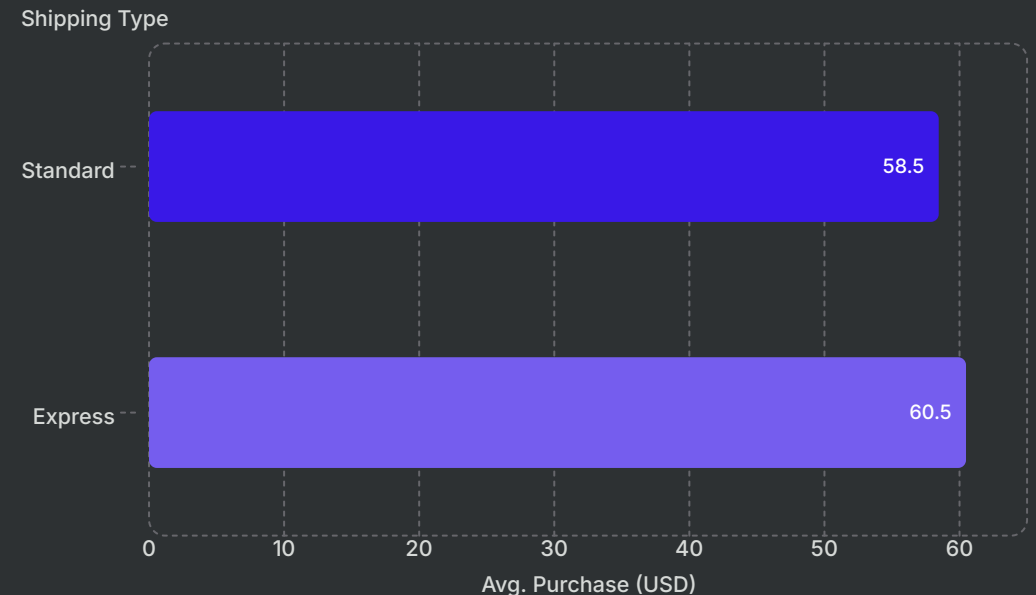


Revenue & Spending Insights

Revenue by Gender

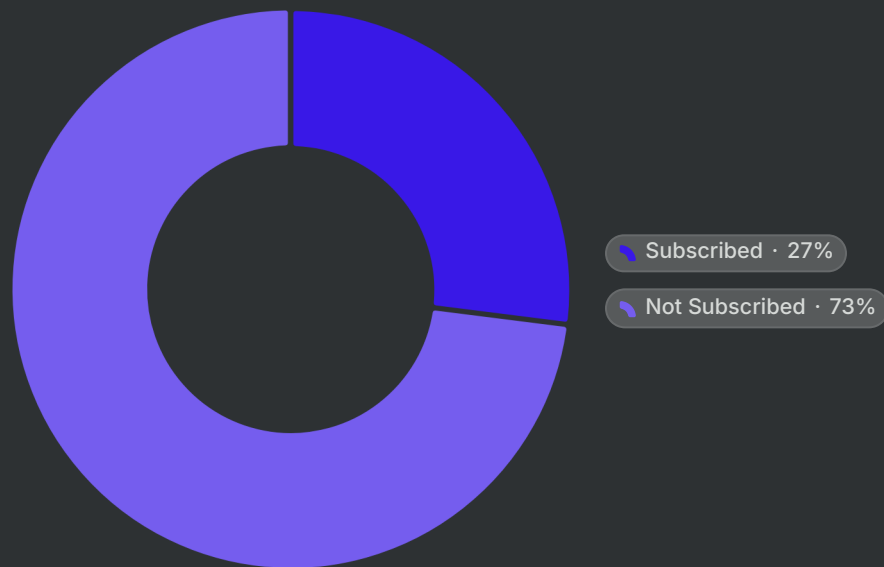


Shipping Type vs. Avg. Spend



Express shipping users spend slightly more on average — a signal for targeted upsell opportunities.

Subscriber vs. Non-Subscriber Behavior

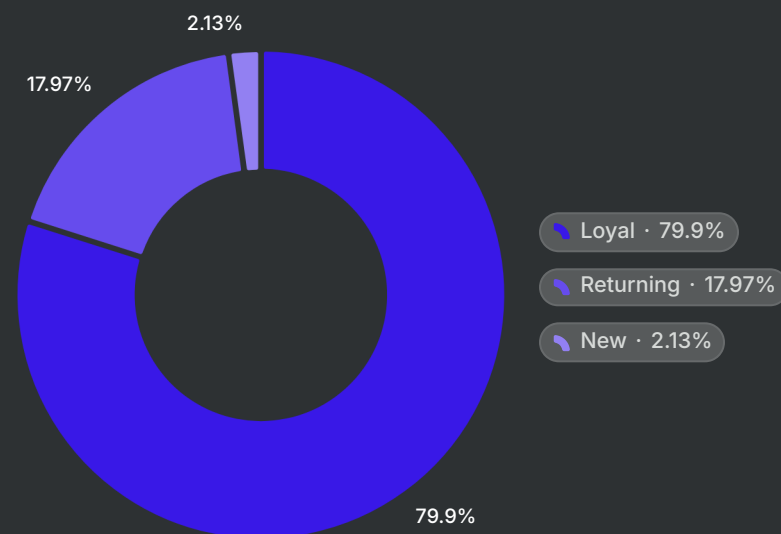


Key Findings

Status	Customers	Avg. Spend	Total Revenue
Subscribed	1,053	\$59.49	\$62,645
Not Subscribed	2,847	\$59.87	\$170,436

Only **27%** of customers subscribe, yet average spend is nearly identical — suggesting strong upside in converting non-subscribers.

Customer Segmentation



Segments by Purchase History

Loyal — 3,116

Majority of the base;
high repeat purchase
count.

Returning — 701

Moderate purchase
history; conversion
opportunity.

New — 83

Small new cohort; focus on onboarding and retention.

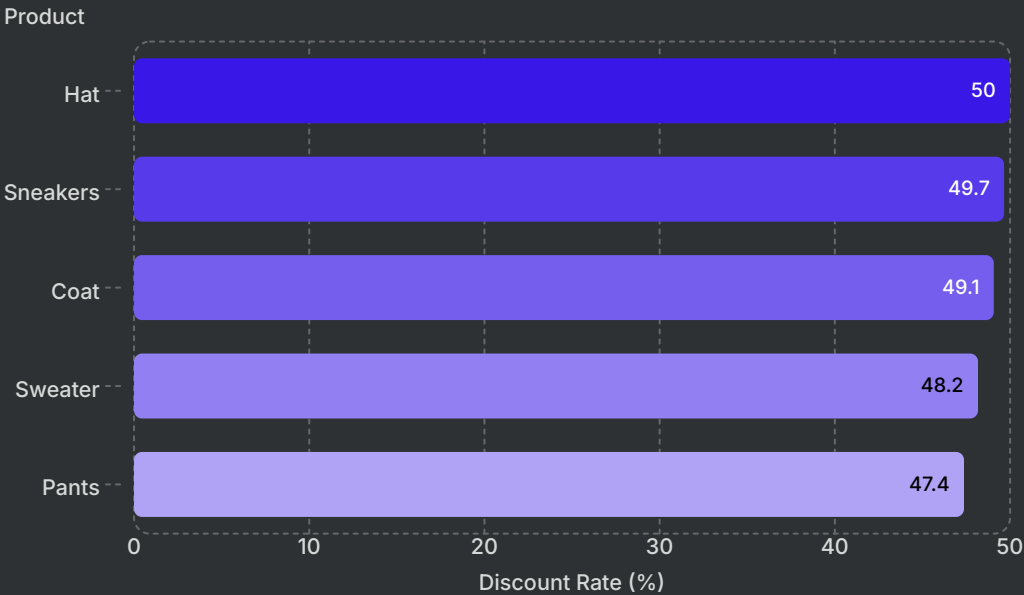
i 958 of 3,476 repeat buyers (>5 purchases) are subscribers.

Top Products & Discount Behavior

Top 5 Products by Rating

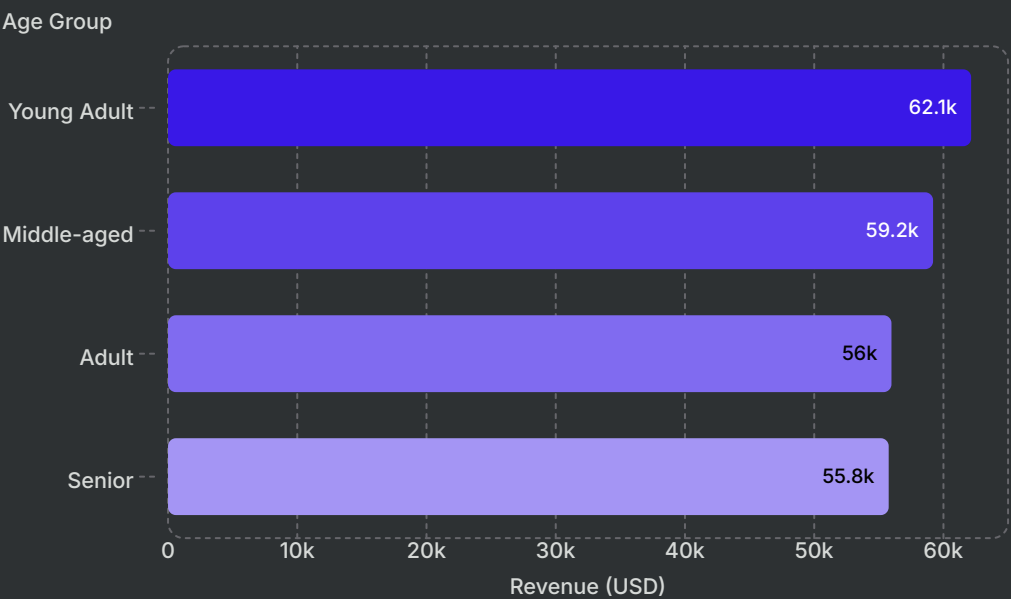
Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products



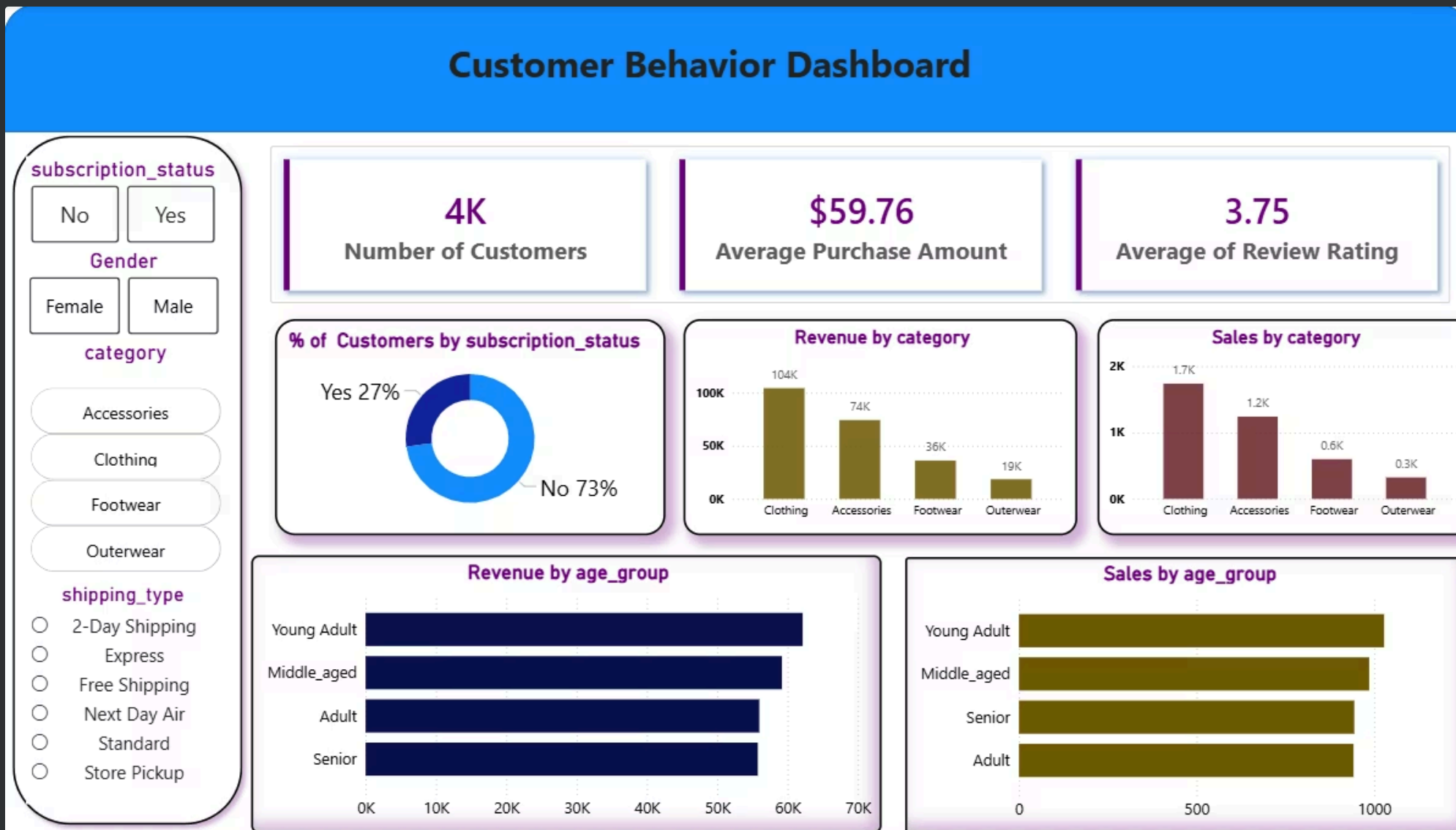
Revenue by Age Group & Top Products by Category

Revenue by Age Group



Top Products per Category

Category	#1	#2	#3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—



Power BI Dashboard

4.K

Customers

\$59.76

Avg. Purchase

3.75

Avg. Rating

Interactive filters by Subscription Status, Gender, Category, and Shipping Type. Visualizations include revenue/sales by category and age group, plus subscription breakdown.

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% of non-subscribers.

Loyalty Programs

Reward repeat buyers to deepen engagement and grow the Loyal segment.

Review Discount Policy

Balance promotional discounts with margin control, especially for Hat, Sneakers, and Coat.

Targeted Marketing

Focus campaigns on Young Adults (highest revenue) and express-shipping users.

